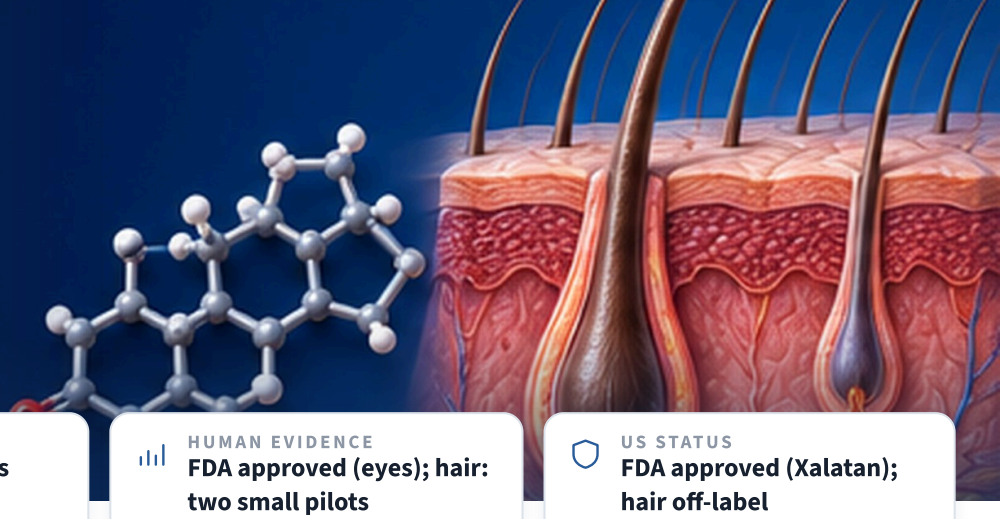
HAIR LOSS
03

Latanoprost

Prostaglandin F2α analogue eye drop, tried on the scalp for hair loss



CLASS	STUDIED FOR	HUMAN EVIDENCE	US STATUS
Prostaglandin analogue	Eye pressure, hair loss	FDA approved (eyes); hair: two small pilots	FDA approved (Xalatan); hair off-label

STUDIED FOR

Eye pressure	Male pattern loss	Female pattern loss
HUMAN	HUMAN	HUMAN

Hair use is off-label. A Phase III trial of a combination solution is recruiting, with no results posted. No compounded spray has been tested.

HOW IT WORKS IN THE BODY

Latanoprost → Active acid → Fluid outflow → Lower eye pressure

- **Prodrug:** corneal esterases turn it into the active acid, per the label
- **Lowers eye pressure** mainly by raising uveoscleral outflow
- **Lash changes** (longer, thicker, more lashes) are a labeled effect
- **On the scalp,** the mechanism is not established

EVIDENCE & WHAT'S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- Eye drop approved in 1996 (Xalatan)
- Hair: two small pilots, 16 men and 29 women
- Phase III combination trial recruiting

NOT YET ESTABLISHED

- Hair effect in larger trials
- Long-term scalp safety
- Scalp dose in people

QUICK PRACTICAL INFO

HALF-LIFE	ROUTES STUDIED
17 minutes (active acid, in plasma)	Eye drops; on the scalp in small trials (off-label)
PHARMACOKINETICS	**SOLD AS**
Ester prodrug hydrolyzed in the cornea; liver β-oxidation	Rx eye drops (Xalatan); compounded scalp sprays
MONITORED IN TRIALS	**SPORT (WADA)**
Hair counts and density (pilots); eye pigment (label)	Not on the 2026 list

REGULATORY

FDA	EMA	WADA
Approved for eye pressure (1996); not hair loss	Authorised nationally for eye pressure	Not on the 2026 list

REFERENCES

1. FDA, Xalatan prescribing information (DailyMed), May 2023
2. Blume-Peytavi et al., J Am Acad Dermatol, latanoprost on the scalp, May 2012
3. Rakowska et al., Pharmacol Rep, latanoprost acid in women, Aug 2026

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